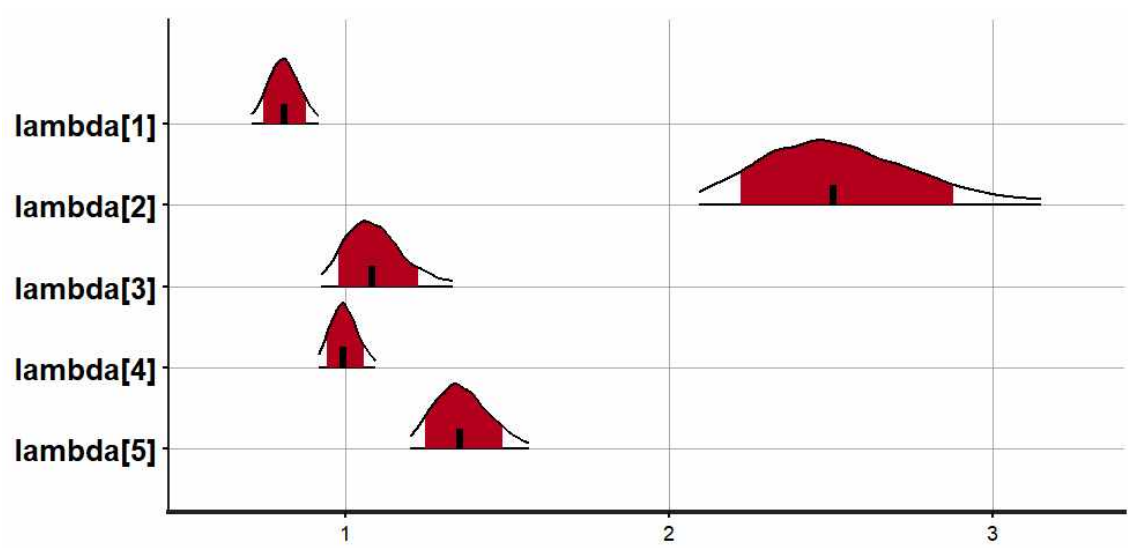
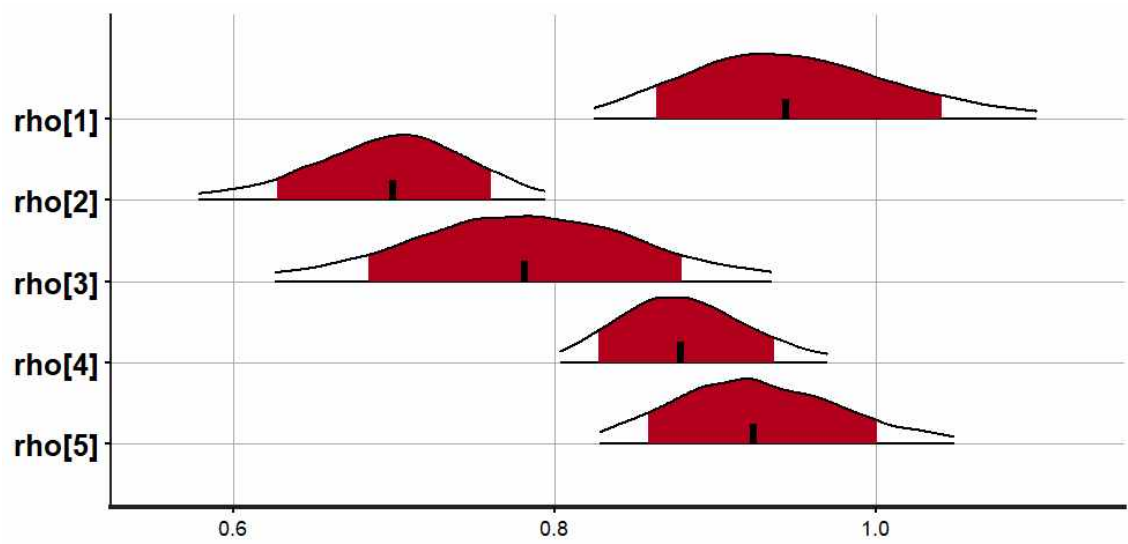
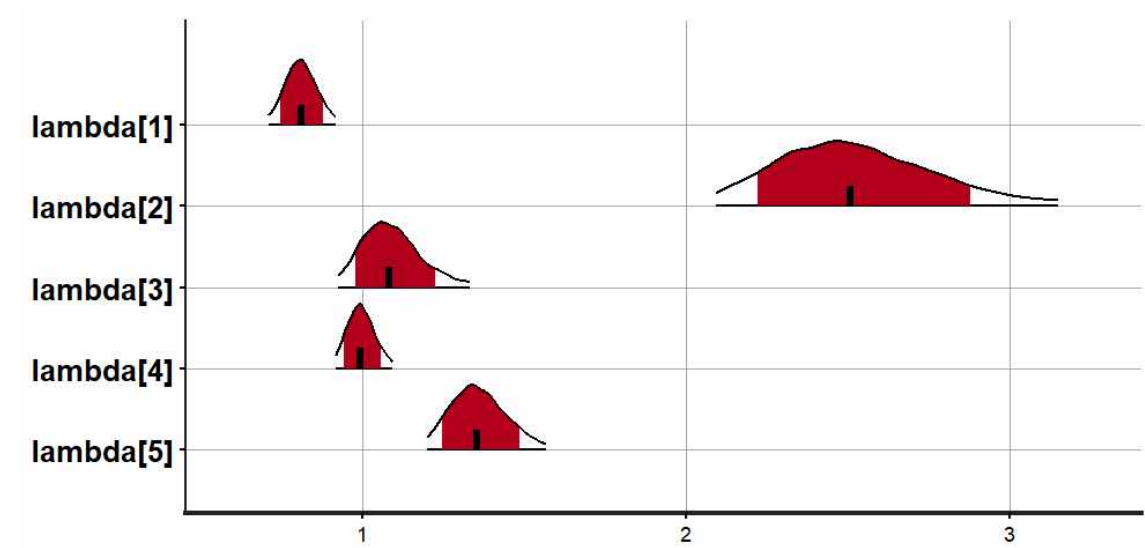


HW5

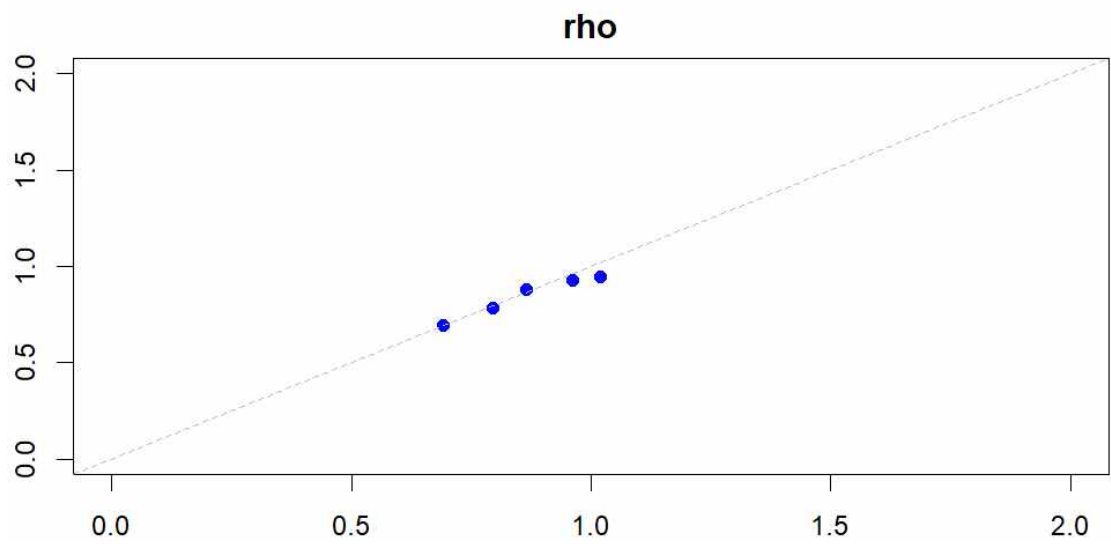
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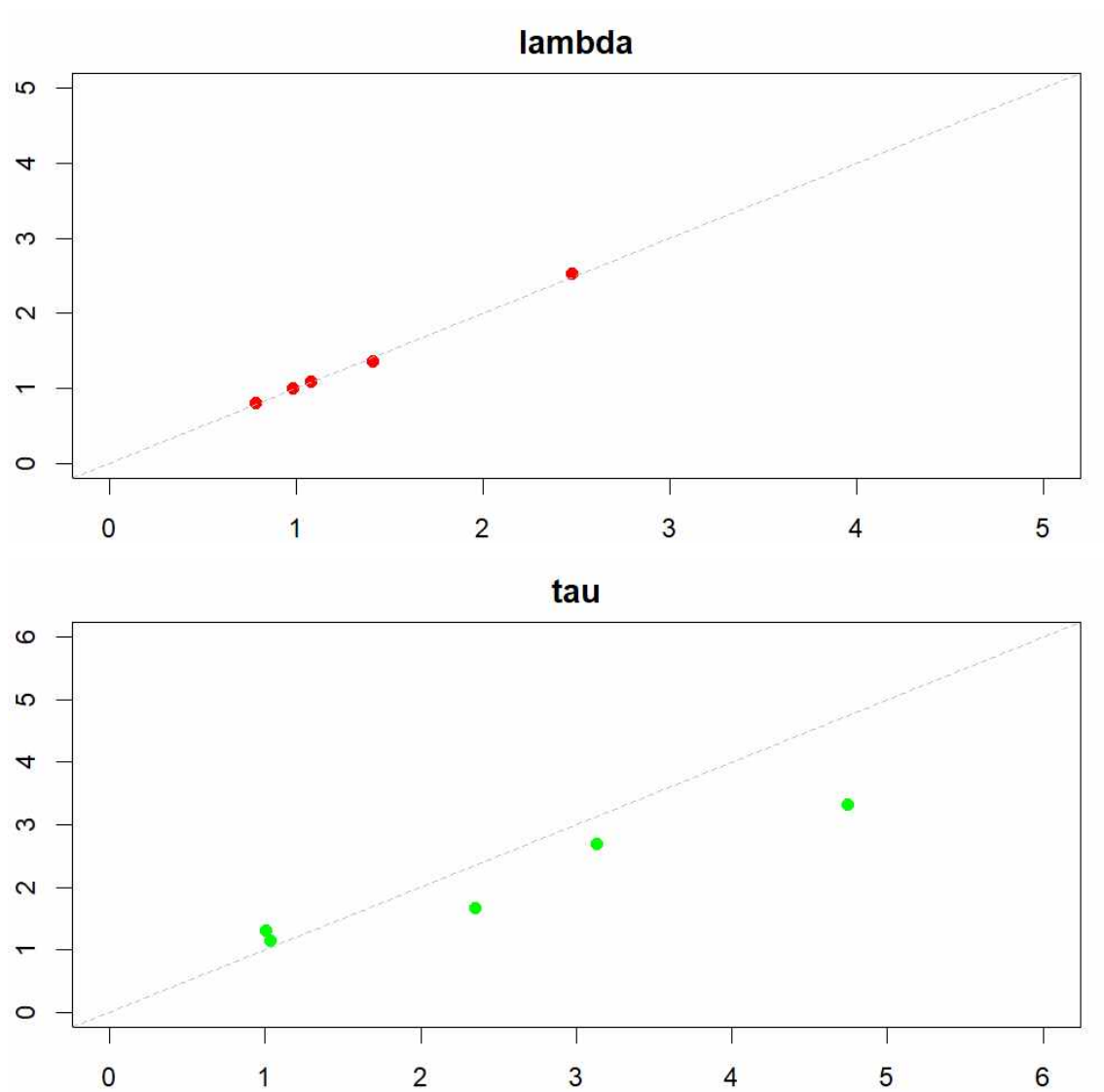
● problem 1





	subjID	mle_rho	mle_lambda	mle_tau	bayes_rho	bayes_lambda	bayes_tau
1	2	1.0199	0.7828	1.0071	0.9489661	0.8096898	1.313048
2	3	0.6927	2.4747	4.7421	0.6962790	2.5331980	3.321189
3	4	0.7944	1.0765	1.0355	0.7818723	1.0926225	1.137539
4	6	0.8656	0.9823	3.1299	0.8806225	0.9963777	2.693549
5	7	0.9616	1.4085	2.3468	0.9282360	1.3607500	1.670165





It shows Shrinkage effect. hBayesDM uses a hierarchical structure. It made individual estimates to be pulled toward the group mean. In subjects 3(in graph number 2) the MLE estimate of 4.742 was shrunk to 3.321.

● problem 2

```
> print(modelcomp_summary)
  Models      AIC      BIC
1 ra_prospect 536.309 580.434
2   ra_noLA 622.449 651.866
3   ra_noRA 557.543 586.960
```

=> In HW 2, we already calculated BIC of three models.

using $\exp(-(BIC_2 - BIC_1)/2)$, calculate BF₁₂. BIC_{prospect} is 580.434, BIC_{noLA} is 651.866. So BF₁₂ is 3.245351e+15.

BF₁₂ is more than 100, so we can conclude ra_prospect (Model 1) is more powerful than ra_noLA (Model 2)

Posterior Odds = BF₁₂ × Prior Odds. And P(M₁)=0.6 and P(M₂)=0.4, So prior odds is 1.5. Because prior odds is P(M₁) / P(M₂).

So, Posterior Odds is calculated by 1.5 X BF₁₂. It is about 4.868 X 10¹⁵

Posterior Model Probability(P(M₁ | D)) is ((Posterior_odds) / (1 + Posterior_odds)). It is about 1 indicating that the data overwhelmingly supports the ra_prospect model.

```
> print(modelcomp_summary)
  Models      AIC      BIC
1   POW1  2.014  2.093
2   POW2  4.014  4.173
3   EXP1  2.314  2.393
4   EXP2  4.107  4.266
5  EXPOW  6.014  6.252
6   HYP1  2.073  2.152
7   HYP2  4.047  4.205
```

Using $\exp(-(BIC_{POW1} - BIC_{models})/2)$. According to BF₁₂, 2 is Models, 1 is POW1.

We get table.

POW2 : 2.829

EXP1 : 1.162

EXP2 : 2.964

EXPOW : 8

HYP1 : 1.03

HYP2 : 2.875

All Bayes Factors were below 10. It means that the differences among the seven models are small

● problem 3

Calculate posterior means of 5 subjects

	subjID	rho	lambda	tau
1	2	1.0308	0.7855	1.0105
2	3	0.6809	2.5262	4.5896
3	4	0.7862	1.1250	1.0208
4	6	0.8689	0.9870	3.1076
5	7	0.9594	1.3844	1.5222

Estimate posterior distributions of all 5 subjects.

All five subjects showed $\rho < 1$. It means risk-averse tendencies. Subject 3 displayed the strongest loss aversion ($\lambda = 2.526$) and the highest inverse temperature ($\tau = 4.590$), suggesting highly consistent and loss-sensitive decision-making. Subject 2 showed $\lambda < 1$, indicating relative insensitivity to losses compared to gain. These results are broadly consistent with Prospect Theory.